Continuous-Time Reinforcement Learning for Portfolio Optimization under Partial Information

A Review of Wu and Li (2024)

Sam Olson Craig Orman

Department of Statistics
Iowa State University

May 4, 2025

Outline

1. Introduction & Motivation
2. Background
3. Methods: The RSEMV Problem & POEMV
4. Results
5. Discussion & Open Problems
6. Conclusion

Why Continuous-Time RL?

- Classical RL operates in **discrete time** — ill-suited for financial markets, which evolve continuously.
- Continuous-time RL reframes decision-making using stochastic processes, diffusion operators, and HJB PDEs.
- Continuous time can can provide unique optimal policy representations unavailable in discrete time.
- The key challenge: how to **explore** without collapsing prematurely to a deterministic policy.

Why Regime Switching?

- Financial markets exhibit pronounced **non-stationarity** (bull and bear markets).
- Regime switching is well-documented as important for long-term investment.
- The regime is **unobservable**: investors can only infer it from prices.
- Classical RL (e.g., EMV) ignores regime structure; its policy is **ignorant of market state**.

Figure: S&P 500 (1992–2022): bull (green) / bear (red).

RL vs. the Classical Approach

Classical approach

1. Estimate model parameters from data.
2. Plug estimates into analytical optimal control.

Problem: Mean return is notoriously hard to estimate — even small errors produce large changes in holdings (*mean blur*). No self-correction mechanism.

RL approach

1. Use filtering to obtain belief state p_t.
2. Learn optimal policy through repeated market interactions.

Advantage: Policy self-corrects from experienced outcomes. Estimation error impacts *filtering only*, not the policy directly.

Paper's Contribution

Wang and Zhou (2020) established the EMV framework for continuous-time mean–variance portfolio selection under full information. Wu and Li (2024) extend this to the partial-information regime-switching setting, yielding the **POEMV algorithm**.

The MV Portfolio Problem

Over a fixed horizon $[0, T]$, minimize terminal wealth variance subject to a mean target:

$$\min_u \ \mathrm{Var}[X_T^u] \quad \text{s.t.} \quad \mathbb{E}[X_T^u] = z, \quad u \text{ self-financed and admissible.}$$

Via a Lagrange multiplier ω, this is equivalent to:

$$\min_u \ \mathbb{E}\big[(X_T^u - \omega)^2\big] - (\omega - z)^2.$$

- **Time-inconsistent**: the variance term prevents a dynamically optimal solution.
- Wu and Li (2024) focus on the **precommitted strategy**, optimal only at $t = 0$.
- The multiplier ω^* is found via the self-correcting rule $\omega \leftarrow \omega - \eta_\omega(X_T - z)$.

Exploratory Stochastic Control

Key idea: replace point-action controls u_t with *distribution-valued* policies $\boldsymbol{\pi}_t \in \mathcal{P}(\mathcal{U})$.

The **exploratory wealth SDE** under stochastic policy $\boldsymbol{\pi}$:

$$d\tilde{X}_t^{\boldsymbol{\pi}} = \sigma\hat{\rho}_t\big(\int u\,\boldsymbol{\pi}(u)\,du\big)\,dt + \sigma\big(\int u\,\boldsymbol{\pi}(u)\,du\big)\,d\hat{B}_t + \sigma\sqrt{\int u^2\boldsymbol{\pi}(u)\,du - \big(\int u\,\boldsymbol{\pi}(u)\,du\big)^2}\,d\bar{B}_t$$

where $\bar{B}$ is an *independent* Brownian motion introduced by randomization.

Entropy-regularized objective (temperature $\Lambda \geq 0$):

$$\min_{\boldsymbol{\pi}}\ \mathbb{E}\left[(X_T^{\boldsymbol{\pi}} - \omega)^2 - \Lambda\int_0^T \mathcal{H}(\boldsymbol{\pi}(\cdot \mid t, X_t, p_t))\,dt\right] - (\omega - z)^2$$

- $\Lambda \to 0$ recovers deterministic optimal control.
- For LQ problems: optimal $\boldsymbol{\pi}^*$ is **Gaussian** with tractable mean and variance.

Regime Switching & The Market Model

The Partial Information Problem

I_t is **unobservable**. The investor observes only prices $\{S_t\}$, so the portfolio must be adapted to the filtration $\mathcal{F}_t^S$.

Risky asset price follows a **regime-switching GBM**:

$$dS_t = S_t(\mu_t\, dt + \sigma\, dB_t)\,, \qquad \mu_t \in \{\mu_1, \mu_2\}$$

where the regime $I_t \in \{1, 2\}$ evolves as a **CTMC** with generator

$$Q = \begin{pmatrix} -\lambda_1 & \lambda_1 \\ \lambda_2 & -\lambda_2 \end{pmatrix}, \qquad \lambda_i > 0.$$

Sharpe ratio in regime i: $\rho_i = (\mu_i - r)/\sigma$.

Wonham Filtering & the Separation Principle

Define the **belief state** $p_t = P(I_t = 1 \mid \mathcal{F}_t^S)$. The **Wonham filter** gives:

$$dp_t = \left[-(\lambda_1 + \lambda_2)p_t + \lambda_2\right]dt + (\rho_1 - \rho_2)\, p_t(1 - p_t)\, d\hat{B}_t$$

where $d\hat{B}_t = dX_t/(\sigma u_t) - \hat{\rho}_t\, dt$ is the **innovation process**
and $\hat{\rho}_t = (\rho_1 - \rho_2)p_t + \rho_2$ is the filtered Sharpe ratio.

Key facts:

- $p = 0$ and $p = 1$ are **entrance boundaries**: p_t stays in $(0, 1)$ in finite time — no boundary conditions needed.
- By the **separation principle**, the partial-information problem reduces to a fully observed HJB in the augmented state (X_t, p_t).
- Wealth dynamics: $dX_t = \sigma u_t(\hat{\rho}_t\, dt + d\hat{B}_t)$ — driven entirely by observables.

The HJB Equation

The optimal value function satisfies the **HJB equation**:

$$v_t + \min_{\boldsymbol{\pi}} \int_{\mathbb{R}} \Big[\tfrac{1}{2}\sigma^2 u^2 v_{xx} + \big[(\rho_2 + (\rho_1 - \rho_2)p)v_x + (\rho_1 - \rho_2)p(1-p)v_{xp}\big] u\sigma + \Lambda \ln \boldsymbol{\pi}(u) \Big] \boldsymbol{\pi}(u)\, du$$

$$+ \tfrac{1}{2}(\rho_1 - \rho_2)^2 p^2 (1-p)^2 v_{pp} + \big[\lambda_2 - (\lambda_1 + \lambda_2)p\big] v_p = 0$$

with terminal condition $v(T, x, p) = (x - \omega)^2 - (\omega - z)^2$.

- Minimizing over $\boldsymbol{\pi}$ yields a **Gaussian optimal policy** — the key structural result.
- The nonlinear PDE is solved via the ansatz:

$$v = (x - \omega)^2 / f(t, p) + g(t, p) - (\omega - z)^2.$$

Value Function & PDE Structure (Theorem 1)

The **optimal value function** takes the semi-analytical form:

$$V^*(t,x,p) = \frac{(x-\omega)^2}{f(t,p)} + g(t,p) - (\omega - z)^2$$

where f and g satisfy **linear parabolic PDEs** in (t,p):

$$f_t + \tfrac{1}{2}(\rho_1-\rho_2)^2 p^2 (1-p)^2 f_{pp} - [\cdots] f_p + (\rho_2 + (\rho_1-\rho_2)p)^2 f = 0, \quad f(T,p) = 1$$
$$g_t + \tfrac{1}{2}(\rho_1-\rho_2)^2 p^2 (1-p)^2 g_{pp} + [\cdots] g_p - \tfrac{\Lambda}{2} \ln \tfrac{\pi \Lambda f}{\sigma^2} = 0, \quad g(T,p) = 0$$

Key properties:

- PDEs are **degenerate**: second-order coefficient vanishes at $p \in \{0,1\}$ — no boundary conditions needed (entrance boundaries).
- Well-posedness via comparison principle; $f \in [1, e^{\bar{\rho}^2(T-t)}]$ with f_p, f_{pp} uniformly bounded.

Optimal Policy: Myopic–Hedging Decomposition (Theorem 2)

The optimal exploration policy is **Gaussian**:

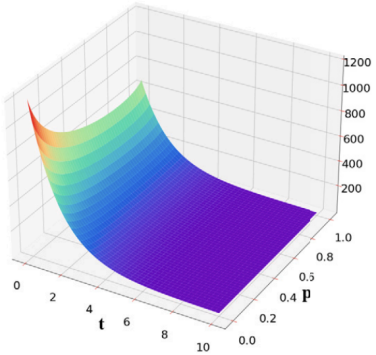
$$\boldsymbol{\pi}^*(\cdot \mid t, x, p) = \mathcal{N}\left(m^*(t,x,p),\ \frac{f(t,p)\Lambda}{2\sigma^2}\right)$$

where the **mean decomposes** as:

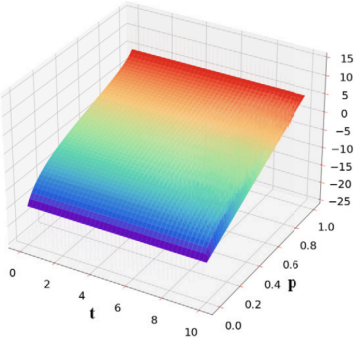
$$m^*(t,x,p) = \underbrace{-\frac{x-\omega}{\sigma}\left[\rho_2 + (\rho_1-\rho_2)p\right]}_{\text{myopic: current filtered Sharpe ratio}} + \underbrace{\frac{x-\omega}{\sigma}(\rho_1-\rho_2)p(1-p)\frac{f_p}{f}}_{\text{hedging: regime transition risk}}$$

- Hedging demand is **largest at** $p_t \approx \frac{1}{2}$ (maximum uncertainty), reducing risky exposure; vanishes when beliefs concentrate near 0 or 1.
- Setting $\rho_1 = \rho_2$ eliminates hedging and recovers the single-regime case.
- Policy variance $\propto \Lambda f(t,p)$ decays as $t \to T$ — exploration naturally winds down.

PDE Surfaces: $f(t,p)$ and $\gamma^*(t,p)$



(a) $f(t,p)$



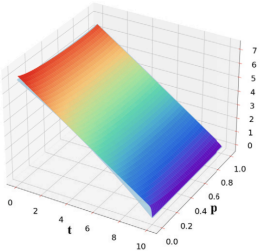
(b) $\gamma^*(t,p)$

- $f(t,p)$ **decreases in** t: exploration variance decays over the horizon.
- $\gamma^*(t,p)$ **increases in** p: invest more in the risky asset as the bull-market probability rises (when $x < \omega$).

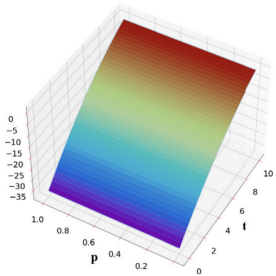
Polynomial Approximation of f and g

Since f and g have no closed form, Wu and Li (2024) approximate:

$$\ln f^{\theta}(t,p) = \sum_{\substack{0\le i\le m\\ 1\le j\le m}} \theta_{i,j}\, p^i (T-t)^j, \qquad g^{\theta}(t,p) = \sum_{\substack{0\le i\le m\\ 1\le j\le m}} \tilde{\theta}_{i,j}\, p^i (T-t)^j$$



(a) $\ln f(t,p)$ (rainbow) vs. OLS fit (light blue)



(b) $g(t,p)$ (rainbow) vs. OLS fit (light blue)

At order $m = 2$: relative MAE = **1.03%** for $\ln f$, **0.06%** for g — justifying $m = 2$ throughout.

The POEMV Algorithm

The parametrized policy is Gaussian with mean and variance driven by f^θ:

$$\boldsymbol{\pi}^\phi(\cdot \mid t, x, p) = \mathcal{N}\left(-(x-\omega)\left[\phi_1\left(p - \frac{f_p^\theta}{f^\theta}p(1-p)\right) + \phi_2\right],\ \Lambda e^{\phi_3} f^\theta(t,p)\right)$$

Each training episode alternates four steps:

1. **Filter:** compute p_{t_k} via discretized Wonham filter.
2. **Critic (PE):** minimize martingale loss $\mathcal{ML}(\theta;\phi)$ by gradient descent on θ.
3. **Actor (PG):** score-function gradient step on ϕ.
4. **Lagrange update** (every 10 iterations): $\omega \leftarrow \omega - \eta_\omega(\bar{X}_T - z)$.

Why polynomial parametrization, not neural networks?

Structure-informed polynomials consistently outperform DDPG and SAC here — fewer parameters, more stable training.

Martingale Loss for Policy Evaluation

For any policy $\boldsymbol{\pi}$, the process

$$M_t^{\pi} = V^{\pi}(t, X_t, p_t) - \Lambda \int_0^t \mathcal{H}(\boldsymbol{\pi}(\cdot \mid s, X_s, p_s))\, ds$$

is a **martingale**. Any nonzero drift reveals a mis-specified value function.

The **martingale loss** (critic objective), discretized over grid $\{t_0, \ldots, t_K\}$:

$$\mathcal{ML}_{\Delta t}(\theta; \phi) := \tfrac{1}{2}\, \overline{\mathbb{E}} \sum_{k=0}^{K-1} \left[h(X_T^{\boldsymbol{\pi}^{\phi}}) - V^{\theta}(t_k, X_{t_k}, p_{t_k}) - \sum_{i=k}^{K-1} \Lambda \mathcal{H}(\boldsymbol{\pi}^{\phi}) \Delta t \right]^2 \Delta t$$

- Unique minimizer coincides with the true value function parameters.
- Avoids estimating second derivatives of V from data (unlike HJB residual methods).
- Same structure as Wang and Zhou (2020) — architecture transfers from full to partial information **with minimal modification**.

Simulation Setup: Calibrated Parameters

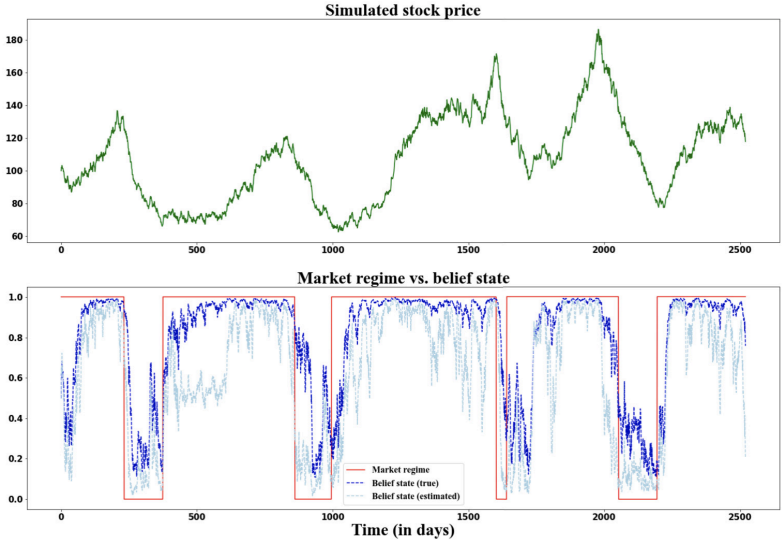
Parameters calibrated from 20 years of daily S&P 500 data (Dai et al. 2010 heuristic):

μ_1	μ_2	σ	λ_1	λ_2
0.25	-0.73	0.22	0.36	2.89

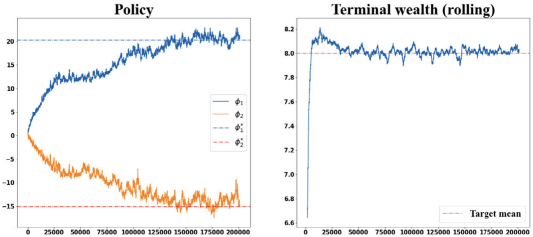
Table: True environment parameters (Table 1 of Wu and Li (2024))

- Horizon: 10 years; $x_0 = 1$, target $z = 8$, $\Delta t = 1/252$, $\Lambda = 1$; 200,000 training episodes.
- **EM algorithm** produces implausible transition rates ($\hat{\lambda}_1 \approx 128$, $\hat{\lambda}_2 \approx 132$) due to low SNR at daily frequency — heuristic approach adopted throughout.

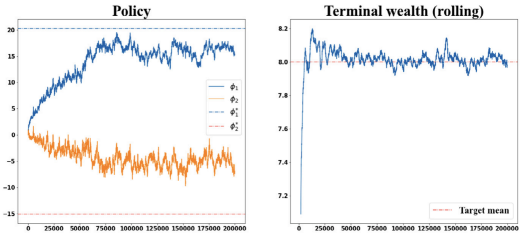
Filtering Quality: Belief State Tracking



Simulation Convergence Results



(a) POEMV with true environment parameters



(b) POEMV with estimated environment parameters

Simulation: Out-of-Sample Performance

Out-of-sample evaluation on 100,000 independent 10-year test paths:

Table: Out-of-sample performance (Table 3 of Wu and Li (2024))

	ϕ_1	ϕ_2	Mean	Variance	Sharpe
Optimal (FD)	20.25	-15.08	7.99	0.067	27.00
POEMV (true)	20.56	-15.05	7.99	0.083	24.26
POEMV (est.)	15.23	-7.29	8.03	0.170	17.05

- With true parameters, POEMV nearly matches the finite-difference optimal.
- With estimated parameters, variance (0.170) remains far smaller than classical alternatives.
- **Dominant gap: filtering error**, not actor–critic learning error.

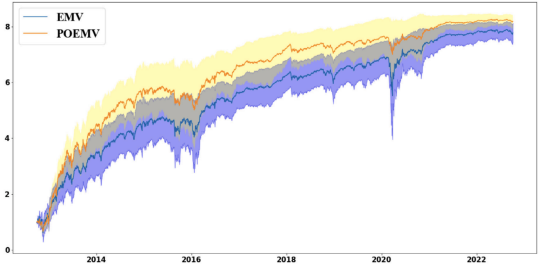
Empirical Study: S&P 500 Setup

- **Data:** 30 years daily S&P 500 + 121 constituent stocks (09/1992–10/2022); 20-year train / 10-year test split.
- **Universal training:** 30 stocks with $\hat{\beta} \in [0.9, 1.1]$ as index proxies; overlapping 10-year blocks $\approx$ 75,000 episodes.
- **Baselines:**
  - **EMV**: same architecture as POEMV, but *ignores regime switching*.
  - **Classical plug-in**: estimate parameters, then apply analytical optimal control.

Why EMV is the key comparison

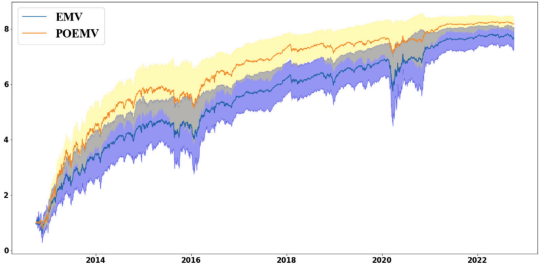
POEMV and EMV share the same architecture — their performance difference isolates the **contribution of regime-awareness alone**.

Empirical Results: Wealth Trajectories



(a) Daily rebalancing

(a) Daily rebalancing



(b) Monthly rebalancing

(b) Monthly rebalancing

POEMV shows faster growth in the bull market and better protection during the **March 2020 market dislocation**.

Empirical Results: Performance Tables

(a) Daily rebalancing

	Mean	Var	Sharpe
POEMV	8.19	**0.224**	**15.38**
EMV	7.79	0.453	10.09
Classical	8.15	1.537	5.77

(b) Monthly rebalancing

	Mean	Var	Sharpe
POEMV	8.19	**0.284**	**13.49**
EMV	7.69	0.597	8.66
Classical	8.22	10.081	2.27

- POEMV variance $\approx \frac{1}{2}$ of EMV (daily); gap $> 10\times$ vs. classical (monthly).
- POEMV also outperforms under leverage constraints $\ell = 2, 3$ (Table 5).

Open Problems

Wu and Li (2024) establish structural results (Theorems 1–2) but **no end-to-end convergence guarantee** exists. Four challenges remain:

1. **Convergence:** Filter error, polynomial approximation error, and stochastic gradient error are not jointly analyzed. A rigorous rate result remains open.
2. **Time consistency:** The separation principle holds for the *precommitted* strategy only. Extending to *equilibrium* strategies requires new theory.
3. **Joint estimation & control:** The current architecture *sequences* filtering and learning. A fully model-free coupling of the Wonham filter and actor–critic updates remains open.
4. **Scalability:** Two regimes, one risky asset. Multi-asset portfolios and richer latent structure interact non-trivially with the filtering and control problems.

Conclusion

1. **Gaussian optimality survives partial information** via the Wonham filter and separation principle — the foundational result enabling POEMV.
2. **The martingale actor–critic transfers cleanly** from full to partial information with minimal modification — the central contribution of Wu and Li (2024).
3. **Regime-awareness materially improves performance:** POEMV halves out-of-sample variance vs. EMV; the gap exceeds an order of magnitude vs. classical approaches under monthly rebalancing.
4. **Filtering error, not learning error, is the dominant bottleneck** — pointing toward joint parameter estimation and control as the most impactful open problem.

References I

- Wang, Haoran and Xun Yu Zhou (2020). "Continuous-Time Mean–Variance Portfolio Selection: A Reinforcement Learning Framework". In: *Mathematical Finance* 30.4, pp. 1273–1308. DOI: 10.1111/mafi.12281.
- Wu, Bo and Lingfei Li (2024). "Reinforcement Learning for Continuous-Time Mean-Variance Portfolio Selection in a Regime-Switching Market". In: *Journal of Economic Dynamics and Control* 158, p. 104787. DOI: 10.1016/j.jedc.2023.104787.

Appendix: POEMV Algorithm (Algorithm 1)

Require: learning rates $\eta_\theta, \eta_\phi, \eta_\omega$; wealth x_0; target z; horizon T; $\Delta t = T/K$; Λ; iterations M; Lagrange update frequency s.

Initialize: Estimate $\tilde{\sigma}, \tilde{\mu}_i, \tilde{\lambda}_i$ from historical data. Set $\theta, \phi = \mathbf{0}$.

For $m = 1$ to M:

1. **Episode:** At each t_k, observe (S_{t_k}, X_{t_k}); filter p_{t_k} via Wonham; sample $u_{t_k} \sim \boldsymbol{\pi}^\phi$.
2. **PE update:** $\theta \leftarrow \theta + \eta_\theta \, \Delta\theta$ (martingale loss gradient).
3. **PG update:** $\phi \leftarrow \phi - \eta_\phi \, \Delta\phi$ (score-function gradient).
4. **If** $(m \bmod s) = 0$: $\omega \leftarrow \omega - \eta_\omega \left(\frac{1}{s} \sum X_T^j - z \right)$.

Appendix: Discretized Wonham Filter

Apply Itô's formula and Euler scheme to $q_t^1 = \log p_t$ and $q_t^2 = \log(1 - p_t)$ to guarantee $p_{t_k} \in (0, 1)$:

$$q_{t_k}^1 = q_{t_{k-1}}^1 + \left[\tilde{\lambda}_2 e^{-q_{t_{k-1}}^1} - (\tilde{\lambda}_1 + \tilde{\lambda}_2) - \tfrac{1}{2}(\tilde{\rho}_1 - \tilde{\rho}_2)^2 (1 - e^{q_{t_{k-1}}^1})^2\right] \Delta t + (\tilde{\rho}_1 - \tilde{\rho}_2)(1 - e^{q_{t_{k-1}}^1}) \Delta \hat{B}_{t_k}$$

Then: $p_{t_k} = e^{q_{t_k}^1} / (e^{q_{t_k}^1} + e^{q_{t_k}^2})$, guaranteed in $(0, 1)$.

Innovation process increment:

$$\Delta \hat{B}_{t_k} = \frac{1}{\tilde{\sigma}} \left[\log \frac{S_{t_k}}{S_{t_{k-1}}} - (\hat{\mu}_{t_{k-1}} - \tilde{\sigma}^2/2)\Delta t\right], \quad \hat{\mu}_{t_{k-1}} = \tilde{\mu}_1 p_{t_{k-1}} + \tilde{\mu}_2 (1 - p_{t_{k-1}})$$

Appendix: Estimation Method Comparison (Table 2)

Table: Heuristic vs. EM algorithm (100 simulated paths; Table 2 of Wu and Li (2024))

	μ_1	μ_2	λ_1	λ_2	σ
True	0.25	−0.73	0.36	2.89	0.22
Heuristic	0.58 (0.10)	−0.69 (0.13)	1.21 (0.42)	2.09 (0.59)	0.22 (0.003)
EM	0.51 (0.65)	−0.21 (0.28)	128.59 (90.14)	132.21 (88.03)	0.25 (0.12)

- EM fails on transition rates — low SNR at daily frequency causes it to identify regime switches from noise.
- Heuristic adopted throughout; both methods estimate σ accurately.

Appendix: Leverage Constraint Results (Table 5)

With leverage cap ℓ and target $z = 4$ (daily rebalancing):

Table: Performance under leverage constraints (Table 5 of Wu and Li (2024))

	$\ell = 2$		$\ell = 3$	
	Variance	Sharpe	Variance	Sharpe
POEMV	**0.690**	**3.23**	**0.691**	**3.39**
EMV	0.766	2.91	0.734	3.10

POEMV outperforms EMV on mean, variance, and Sharpe in both cases — regime-awareness is beneficial even under leverage constraints.